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QNFO 100-Year Paradigm Forecast v2.0

Calibrated Bayesian Cascade Model with Scenario Testing (2026–2126+)

Authors: QNFO Research Collective **Date:** 2026-07-16 **Version:** 2.0 (supersedes v1.0.0 at Zenodo 10.5281/zenodo.21366522) **License:** CC-BY-4.0

What This Paper Is (And Is Not)

This paper IS: The substantive QNFO 100-Year Paradigm Forecast — what we predict will happen across 4 eras of quantum computing and fundamental physics, with calibrated probabilities, actionable signposts, and strategic implications.

This paper is NOT: A methodology paper. The CFPE methodology (Bayesian cascade model, adversarial red-teaming, evidence grading, back-casting calibration) is documented separately in the CFPE Methodology v2.0 paper (Zenodo: pending publication).

If you want to understand HOW we arrived at these numbers, read the methodology paper. **If you want to know WHAT we forecast and WHY it matters**, read this paper.

Executive Summary

We forecast a 100-year landscape of quantum computing and fundamental physics across four eras (2026–2126+). The forecast was generated by the Cascading Bayesian Foresight Engine (CFPE) methodology, calibrated through historical back-casting (1976→2026, 1950→2000, 1926→1976), and stress-tested from 8 independent attack vectors.

The headline finding: Gate-based quantum computing will continue scaling incrementally for the next decade (82% probability). Room-temperature quantum computing via NV-diamond centers is the most actionable near-term wildcard (15% probability). The CMB-S4 experiment (results expected ~2035) is the single most important empirical resolution across the entire forecast — it

will either validate or falsify the hydrodynamic foundations of quantum mechanics. Geometric derivation of fundamental constants faces the strongest headwinds (57% pessimistic).

Full cascade probability (all 4 eras as paradigm shifts): 6.5% [4–9%].

ERA 1: The Gate-Based Computing Decade (2026–2036)

What We Forecast

Pathway	Probability	Description
Continuity Baseline	**82%**	Gate-based quantum computers continue scaling at ~10× per 4 years. Surface codes improve. Active QEC manages thermal load. 5,000-10,000 physical qubits by 2036. No paradigm shift — but steady progress.
FCI Breakthrough	**8%**	Fractional Chern insulator braiding interferometry achieves intrinsic fault tolerance. Logical qubits operate above 1 Kelvin. The dilution refrigerator becomes optional. Active QEC is replaced by passive topological protection.
Stagnation	**10%**	The thermal wall bites: at 5,000 qubits, dilution refrigerators cannot maintain base temperature during QEC cycles. Error rates climb with qubit count. The "quantum winter" narrative emerges.

Why This Matters

- **If you're a policymaker:** The CRQC (cryptographically-relevant quantum computer) timeline is 2035-2040, not 2030-2033. PQC migration should proceed on a risk-managed timeline, not a panic-driven one.
- **If you're an investor:** Gate-based QC companies (Google, IBM, IonQ) will continue attracting funding through 2028-2030. The real inflection point is whether someone demonstrates >5,000 high-fidelity qubits without thermal issues. If not, expect a funding correction.
- **If you're a researcher:** NV-diamond centers are the most actionable near-term alternative to gate-based QC. $T > 1$ second at 300K has been demonstrated in isotopically purified diamond. Two-qubit gates at room temperature are the next milestone.

What to Monitor

Signpost	Check Date	What It Means
>5,000 physical qubits with >99.5% two-qubit fidelity	2028	Gate-based scaling is on track — continuity baseline confirmed
Thermal wall observed (cooling failure at >5k qubits)	2029	Era 1 pessimistic scenario — expect funding reallocation
FCI anyon braiding confirmed with >99% fidelity	2029	Era 1 optimistic scenario — FCI becomes viable alternative
NV center two-qubit gate at 300K with >99.9% fidelity	2030	Wildcard validated — room-temperature QC is viable via solid-state

How We Got These Numbers

The FCI optimistic probability dropped from 25% (raw CFPE) to 8% (calibrated) after: - Evidence grading: 0% DIRECT experimental evidence for FCI coherence above millikelvin - Adversarial red-team: Bruhat-Tits trees lack physical realizability mechanism in FCI systems - Fault-tree fragility: 7 AND-gated conditions must ALL succeed for FCI breakthrough - Back-casting: Technological breakthroughs in physics are rarer than the model's raw priors suggest - NV-center wildcard leakage: 15% of probability mass diverted to solid-state room-temperature alternatives

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ERA 2: The Room-Temperature Computing Era (2036–2046)

What We Forecast

Pathway	Probability	Description
Continuity (Gate-Based QC wins)	**64%**	Gate-based quantum computers with cryogenic cooling remain the dominant paradigm. Commercial advantage demonstrated in pharmaceutical and materials science applications. Bio-spintronics is a research footnote.
Bio-Spintronics Breakthrough	**15%**	A controllable square root of SWAP radical-pair gate is demonstrated at 310 Kelvin in a synthetic biomimetic channel. The "viscosity paradox" is resolved: viscoelastic relaxation time matching spin precession period enables biological gate operations.

Pathway	Probability	Description
Ceiling	**21%**	The witness-vs-gate gap proves fundamental: biological systems demonstrate passive quantum coherence (femtosecond-picosecond timescales) but cannot support ACTIVE gate manipulation. Every external control field destroys the coherence it attempts to measure.
Wildcard: NV-Diamond Centers	**15%**	Solid-state spin qubits (NV centers in diamond) achieve room-temperature two-qubit gates with >99.9% fidelity — bypassing the biological pathway entirely. The QNFO insight that "room-temperature coherence is possible" was correct; the extrapolation that "biological substrates are necessary" was wrong.

Why This Matters

- **If you're a technologist:** Room-temperature quantum computing is the holy grail — it eliminates the dilution refrigerator supply chain, reduces per-qubit cost by 100-1000×, and enables deployment outside specialized laboratory environments. The race is between biological substrates (bio-spintronics) and solid-state substrates (NV centers, silicon spin qubits). Solid-state is currently ahead.
- **If you're a biologist:** Cryptochrome magnetoreception remains the most empirically supported quantum biology phenomenon. Understanding how nature achieves spin coherence at 310 Kelvin has implications beyond computing — for magnetic sensing, medical diagnostics, and understanding the evolutionary origins of quantum effects in biology.
- **If you're a strategist:** Era 2 is the highest-leverage era. P(Era 3 Optimistic | Era 2 Optimistic) = 42% (calibrated, down from 85% raw). If room-temperature QC is achieved via ANY pathway, the experimental conditions for testing hydrodynamic quantum foundations are created. Era 2 is the gateway to Eras 3-4.

What to Monitor

Signpost	Check Date	What It Means
Synthetic cryptochrome-mimetic channel: spin coherence >10 ms at 310K	2030	Bio-spintronics on track

Signpost	Check Date	What It Means
Systematic review shows biomimetic coherence ceiling at ~100 microsecond	2032	Bio-spintronics ceiling confirmed — gate operations unlikely
NV center T $>$ 1 second at 300K (already demonstrated 2024 — monitor for gate demonstrations)	2028	NV center wildcard path accelerating
First room-temperature two-qubit gate (any substrate) with >99.9% fidelity	2032	The room-temperature QC era has begun — regardless of substrate

How We Got These Numbers

The bio-spintronics probability dropped from 33.8% (raw) to 15% (calibrated) after: - **The witness-vs-gate gap:** Every expert panel independently identified this as the strongest counterargument. Biological systems witness quantum effects; they don't manipulate them computationally. - **Back-casting calibration:** The 1976→2026 back-cast showed that the CFPE overpredicted speculative technologies (cold fusion, flying cars) and underpredicted engineering-driven breakthroughs (PCs, internet). Bio-spintronics is in the "speculative" category. - **NV center competition:** The wildcard pathway absorbs 15% of probability mass from bio-spintronics. Solid-state room-temperature QC is a competing hypothesis with better near-term evidence.

ERA 3: The Foundations of Physics (2046–2076)

What We Forecast

Pathway	Probability	Description
Standard Quantum Mechanics persists	**47%**	Quantum mechanics remains the correct framework. CMB anomalies are explained within LambdaCDM. Superdeterminism, many-worlds, and objective collapse models compete as interpretations but none achieves consensus empirical distinction from standard QM.

Pathway	Probability	Description
Hydrodynamic Foundations validated	**33%**	The CMB-S4 experiment (or successor) detects log-periodic oscillations at >5 sigma significance. The fundamental scaling ratio is measured at q approximately e (2.718) — consistent with the Syntactic Token Calculus prediction. This provides the first empirical anchor for the hydrodynamic reformulation of quantum mechanics. Superdeterminism transitions from "fringe interpretation" to "alternative framework with empirical support."
Null Results	**35%**	CMB-S4 returns a null result at >5 sigma exclusion. The STC prediction is falsified. The hydrodynamic foundations program is retrospectively understood as an elegant mathematical framework that makes no contact with experiment — "a beautiful theory that nature declined to implement."
Wildcard: CSL Objective Collapse wins	**12%**	Objective collapse models (Continuous Spontaneous Localization, Diosi-Penrose) provide the first direct experimental evidence for physical wavefunction collapse. A tabletop experiment detects spontaneous radiation emission from a macroscopic superposition. The "collapse is physical" insight from QNFO was correct; the hydrodynamic mechanism was wrong.

Why This Matters

- **This is the most philosophically significant era.** If superdeterminism or CSL is confirmed, it rewrites the foundations of quantum mechanics — a paradigm shift comparable to relativity or quantum mechanics itself. The philosophical implications (determinism vs indeterminism, locality vs non-locality, the nature of measurement) would transform our understanding of reality.

- **The CMB prediction is the single most important empirical anchor in the entire 100-year forecast.** CMB-S4 (or the Simons Observatory) will either detect or exclude log-periodic oscillations by approximately 2035. This one result determines whether Era 3 has any empirical content at all. Everything else in Eras 3-4 is downstream of this measurement.
- **The generational turnover problem:** Even if superdeterminism is confirmed in 2035, institutional acceptance will take 20-30 years. The physicists who would need to accept it are currently alive and trained in standard QM. Acceptance tracks retirement, not evidence.

What to Monitor

Signpost	Check Date	What It Means
CMB-S4 detects log-periodic oscillations at >5 sigma, q approximately e	~2035	Era 3 validated — hydrodynamic foundations have empirical anchor
CMB-S4 null result at >5 sigma exclusion	~2035	Era 3 collapsed — superdeterminism has no empirical content
Tabletop experiment detects CSL-predicted spontaneous radiation at >5 sigma	~2032	CSL wildcard pathway validated — objective collapse is physical
>80% of physicists classify superdeterminism as "unfalsifiable in practice" (survey)	~2050	Superdeterminism rejected as unfalsifiable even if logically coherent

How We Got These Numbers

The hydrodynamic foundations probability dropped from 38.3% (raw) to 33% (calibrated) after: - **Formal mathematical audit:** The Bridge Theorem requires a physically unrealizable condition (diagonal clock-rest interaction Hamiltonian). The dynamical "ultrametricity as attractor" claim is untested. - **Reference class anchoring:** Fundamental physics paradigm shifts occur at approximately 0.3 per decade. A single candidate in a single decade should have prior less than or equal to 0.10. - **However:** The CMB prediction is specific, testable, and has a defined resolution window. This is RARE in quantum foundations. Most interpretations make zero testable predictions. The fact that superdeterminism makes ONE testable prediction justifies a higher probability than other interpretations.

ERA 4: The Constants of Nature (2076–2126+)

What We Forecast

Pathway	Probability	Description
Eddington's Ghost (Pessimistic)	**57%**	No geometric derivation of any Standard Model coupling constant achieves consensus acceptance by 2126. The renormalization group flow problem proves fundamental: any geometric derivation must either reproduce the full running of couplings or explain why a specific scale is geometrically privileged. Neither is achieved. The adelic framework produces elegant mathematics with no physical predictions distinguishable from standard physics. "Geometric constant derivation" joins "perpetual motion" and "cold fusion" as historically significant but scientifically failed research programs.
Landscape/Anthropic persists	**45%**	Standard Model constants remain unexplained free parameters within the string landscape. The anthropic principle or cosmological natural selection provides the dominant (if experimentally unverified) explanatory framework. No geometric derivation emerges, but the problem is absorbed into a larger framework.
Geometric breakthrough	**13%**	The fine-structure constant alpha (approximately 1/137.036) is derived from pure geometry — not through pi/phi numerology but through amplituhedron volume in momentum twistor space. The positive Grassmannian geometry of scattering amplitudes provides the mathematical structure. The QNFO insight that "constants have a geometric origin" was correct; the proposed mechanism (pi/phi ratios) was not the right one.

Pathway	Probability	Description
Wildcard: String Landscape absorbs	**8%**	The string landscape measure is proven to be determined by Calabi-Yau topological invariants. Our universe's constants are NOT random draws from 10^500 vacua — they are determined by the geometry of the compactification manifold. The QNFO program was right about geometric origins but wrong about the scale: the relevant geometry is 6-dimensional Calabi-Yau spaces, not pi/phi ratios at the electroweak scale.

Why This Matters

- **This era is the most speculative and least actionable.** The reference class (Eddington 1929 to present) is a century of failure. No geometric derivation has ever achieved consensus acceptance. The prior probability of any specific candidate succeeding in this domain should be very low — which is what our calibrated model reflects (13% optimistic, 57% pessimistic).
- **The value is in asking the right question, not predicting the right answer.** The QNFO program correctly identified that "where do fundamental constants come from?" is one of the deepest open questions in physics. Even if the proposed mechanism (pi/phi geometry) is wrong, the question itself is worth a century of investigation.
- **If ANY geometric derivation succeeds, it would be the most significant theoretical physics breakthrough since the Standard Model.** The practical implications are zero (it wouldn't change any engineering or technology) but the intellectual implications are immense — it would suggest that the laws of physics are determined by mathematical necessity, not contingent historical accident.

What to Monitor

Signpost	Check Date	What It Means
Any SM coupling constant derived from pure geometry to <1% precision	~2045	Era 4 optimistic — the geometric program has succeeded
No-go theorem for geometric derivation of running couplings proven	~2060	Era 4 pessimistic confirmed — geometric derivation is impossible in principle
Adelic framework produces zero experimentally verified predictions by 2100	~2100	Adelic program classified as "mathematical physics with no physical content"

How We Got These Numbers

The geometric constants probability dropped from 31.0% (raw) to 13% (calibrated) after: - **The Eddington reference class is devastating:** A century of attempts with zero successes. No derivation exists. - **The cross-ratio reframing of alpha is mathematically correct but physically circular:** The classical electron radius and Compton wavelength are DEFINED using alpha. Expressing alpha as their ratio is algebra, not derivation. - **Formal mathematical audit:** The adelic framework makes zero predictions that differ from standard physics. If p-adic completions predict the same observables as Archimedean physics, the claim is unfalsifiable. - **Back-casting:** The 1926→1976 back-cast showed the CFPE overpredicted unified field theory (0.25 prior) — Einstein spent 30 years failing. Theoretical physics paradigm shifts are much rarer than the raw model suggests.

Full Cascade Summary

Era	Horizon	Domain	P(Paradigm Shift)	Dominant Scenario
Era 1	2026–2036	FCI / Topological QC	8%	**Continuity (82%)**
Era 2	2036–2046	Bio-Spintronics / NV Centers	15% (bio) + 15% (NV)	**Continuity (64%)**
Era 3	2046–2076	Hydrodynamic Foundations	33%	**Standard QM (47%)**
Era 4	2076–2126+	Geometric Constants	13%	**Eddington's Ghost (57%)**

P(All 4 eras as paradigm shifts): 6.5% [4–9% one-sigma]

P(At least one paradigm shift in any era): ~65%

P(Two or more paradigm shifts): ~25%

P(Three or more paradigm shifts): ~5%

Strategic Implications

For QNFO Strategy

1. **The QNFO program is asking the right questions.** The program correctly identified the thermal wall problem (Era 1), the possibility of room-temperature coherence (Era 2), the need for testable quantum foundations predictions (Era 3), and the geometric origin of fundamental constants (Era 4). Even where the proposed mechanisms are likely wrong, the questions are right.

2. **NV-diamond centers are the priority wildcard.** Room-temperature QC is achievable via solid-state spin qubits — the biological pathway is elegant but harder. QNFO should track NV-center literature monthly.

3. **The CMB prediction is the linchpin.** CMB-S4 results (~2035) will determine whether Eras 3-4 have any empirical content. QNFO should actively engage with CMB-S4 and Simons Observatory collaborations.

4. **Institutional inertia is the binding constraint.** Even if every physical prediction is correct, institutional acceptance adds 15-50 years per era. The forecast timeline (2026-2126) is physics-optimistic. The calibrated timeline with institutional inertia is approximately 2040-2176.

For External Audiences

1. **Policymakers:** CRQC timeline is 2035-2040, not 2030-2033. PQC migration should be risk-managed, not panic-driven. Prepare for the scenario where CRQC never arrives (~5-10% probability).

2. **Investors:** Gate-based QC will continue attracting funding through 2028-2030. The inflection point is whether thermal management scales. Monitor NV-center progress as the most actionable alternative.

3. **Researchers:** The most impactful empirical resolution across the entire 100-year forecast is the CMB-S4 measurement of log-periodic oscillations. This one result determines the credibility of all Era 3-4 claims.

What Would Change This Forecast

Observation	Impact	Cascade EV Shift
CMB-S4 detects log-periodic oscillations at >5 sigma	Eras 3-4 EV up 3-5x	+15-20% cascade EV
CMB-S4 null result at >5 sigma exclusion	Eras 3-4 collapse	-4-5% cascade EV
NV center two-qubit gate at 300K, >99.9% fidelity	Era 2 NV pathway confirmed	+2-3% cascade EV
Gate-based QC exceeds 10,000 qubits without thermal issues	Era 1 FCI thesis weakened	-1-2% cascade EV
Any SM coupling constant derived from pure geometry, <1% precision	Era 4 validated	+5-10% cascade EV
FCI anyon braiding with >99% fidelity demonstrated	Era 1 FCI thesis strengthened	+1-2% cascade EV

Methodology

This forecast was generated by the Cascading Bayesian Foresight Engine (CFPE) methodology, calibrated through historical back-casting and stress-tested from 8 independent attack vectors. Full methodology documentation is available in the CFPE Methodology v2.0 paper (Zenodo: pending publication).

Attack vectors applied to this forecast: 1. Adversarial red-teaming (0% fresh survival rate vs 100% original) 2. Evidence strength grading (0% DIRECT, 20% INDIRECT, 60% THEORETICAL, 20% SPECULATIVE) 3. Fault-tree decomposition (16 independently testable leaf nodes for Era 1) 4. Counterfactual pathway generation (12 futures across 4 eras) 5. Sensitivity expansion (+/- 30% priors, edge removal, impact weight sweep) 6. Comparative forecasting (naive extrapolation vs CFPE model) 7. Formal mathematical auditing (7 claims audited, 5 human-review flags) 8. Expert panel simulation (5 schools of thought, convergent critique)

Back-casting calibration baselines: - 1976 to 2026: Computing revolution (PC, internet, AI, QC) — CFPE underpredicted by 5-10x - 1950 to 2000: Plate tectonics, molecular biology, space-flight — CFPE underpredicted by 5-10x - 1926 to 1976: Nuclear physics, computers, radar — CFPE underpredicted engineering shifts, overpredicted theoretical shifts

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Generated: 2026-07-16 | QNFO 100-Year Paradigm Forecast v2.0